CSI247:
Data Structures

Lecture 01 – Course Introduction

Department of Computer Science
B. Gopolang (247-275)

Learning Objectives

In this lecture we will discuss:

- Course objectives, overview
  - The various topics to be covered in the course
  - Prescribed Textbook
- Course Admin
  - Pre-requisites
  - Timetable
  - Lab registration
  - Assessments
  - Programming environment
- Introduction to Data Structures (DS)

1. Course Objectives

- In this course, you will:
  - Learn about and implement some of the common data structures.
  - Understand costs and benefits of each data structure.

2. Course Learning outcomes

- After completion of the course, you should be able to:

1. Describe common applications for each data structure discussed in the course
2. Implement user-defined data structures using a high-level language (e.g. Java)
3. Choose an appropriate data structure for a given problem
4. Write programs using the discussed data structures
5. Perform worst-case runtime analysis on data structures and simple algorithms

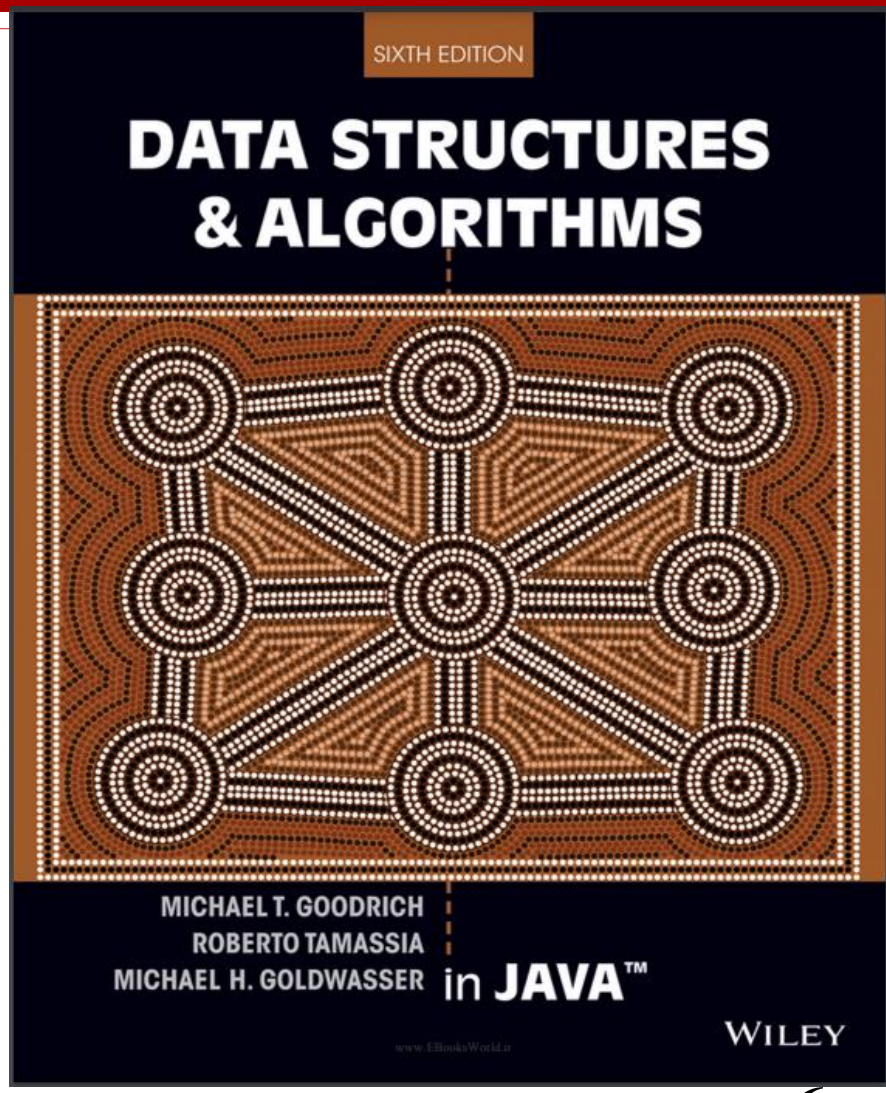
3. Course outline

- Topics will include, but not limited to, the following:
  - Review of Java programming basics
  - Searching and sorting algorithms
    - Using arrays
  - Recursion
  - Time complexity
  - Object oriented programming
  - Use and implementation of common Data Structures:
    - Arrays, Array Lists, Linked lists, Stacks & Queues
    - Trees, Graphs & hash tables (time permitting)

4. Course material

- Lecture notes
  - Handouts (pdfs) - Moodle

- Text book:
  - "Data Structures & Algorithms in Java" by Goodrich, Tamassia & Goldwasser, 6th Ed, 2014



5. Pre-requisites

- Courses:
  - CSI142 (& CSI141) or ISS112: >= 50%
  - **Required basic knowledge**:
    - Java Programming basics
      - Programming structures (loops & selection)
      - File I/O
      - User-defined classes, objects & methods
      - Arrays

6. Timetable

- **Lectures**
  - M, W & F : 7am – 8am, 252/001
- **Lab session**
  - R, 7-9, 9-11am or 11-1pm
    - Pick ONE
    - Lab registration: online (CS Moodle)
      - Changes NOT ALLOWED
    - Room allocation: after lab registration
- **Attendance**
  - Compulsory for ALL (lectures, labs & tutorials!)

7. Assessments

- Final mark:
  - CA **(50%)** + Ex **(50%)**

  a) CA:
    - 4 quizzes (20%)
    - 2 Tests (30%)
      - Date, time and venue for all: TBA

  b) Final Exam: Refer to UB Exam timetable

- **Important:**
  - ALWAYS bring **your <u>valid</u> UB student ID** to these

8. Useful tools/resources

a) CS Moodle
- Handouts (pdfs). Print and bring to class
- Lab exercises
- Announcements

b) Visual Studio Code:
- Programming environment

c) WhatsApp:
- For quick and easy communication
  - Link on CS Moodle

Introduction to Data Structures

Introduction

- Recap: A computer program?
  - Implementation of an algorithm
    - Well-defined set of instructions for solving a specific problem, in a finite amount of time
    - Using a programming language

- What happens during algorithm design?
  - Determine correct logic – processing steps to follow
  - Is that all?

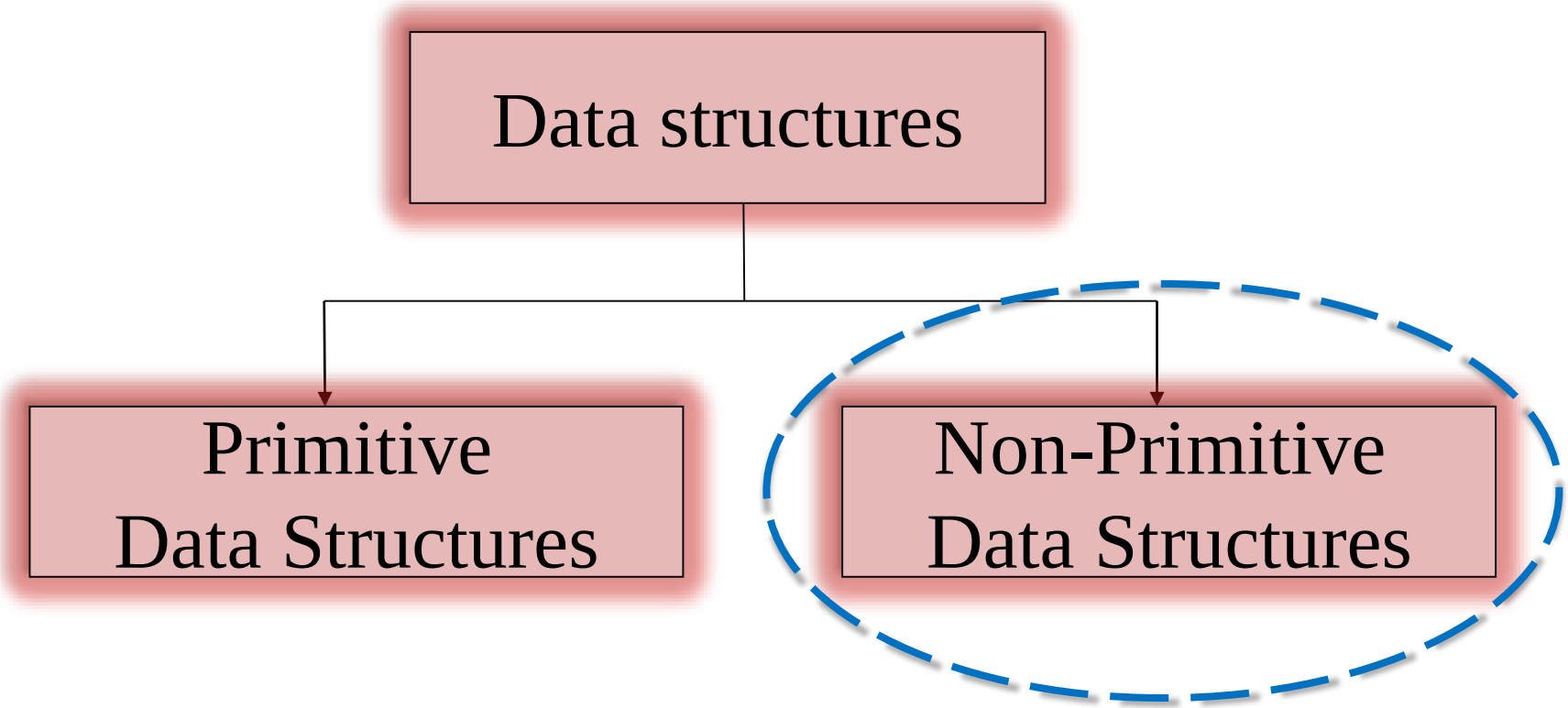
- Answer:
  - No,
  - algorithm alone IS NOT ENOUGH
    - Programs normally manipulate some data!

Program = Algorithm + **Data Structure**

- What is a Data Structure (DS)?
  - A way of **organizing data** in a computer's memory
    - Provides data manipulation framework:
      - Offers guidance on operations allowed on stored data (direct vs sequential access, deletions, etc)
    - Different types available
      - E.g.: array, stack, etc (later !)
  - Note:
    - Choice of data structure affect program design
      - Determines an algorithm's performance

Classification of Data Structures

- 2 broad categories:



UNIVERSITY of BOTSWANA

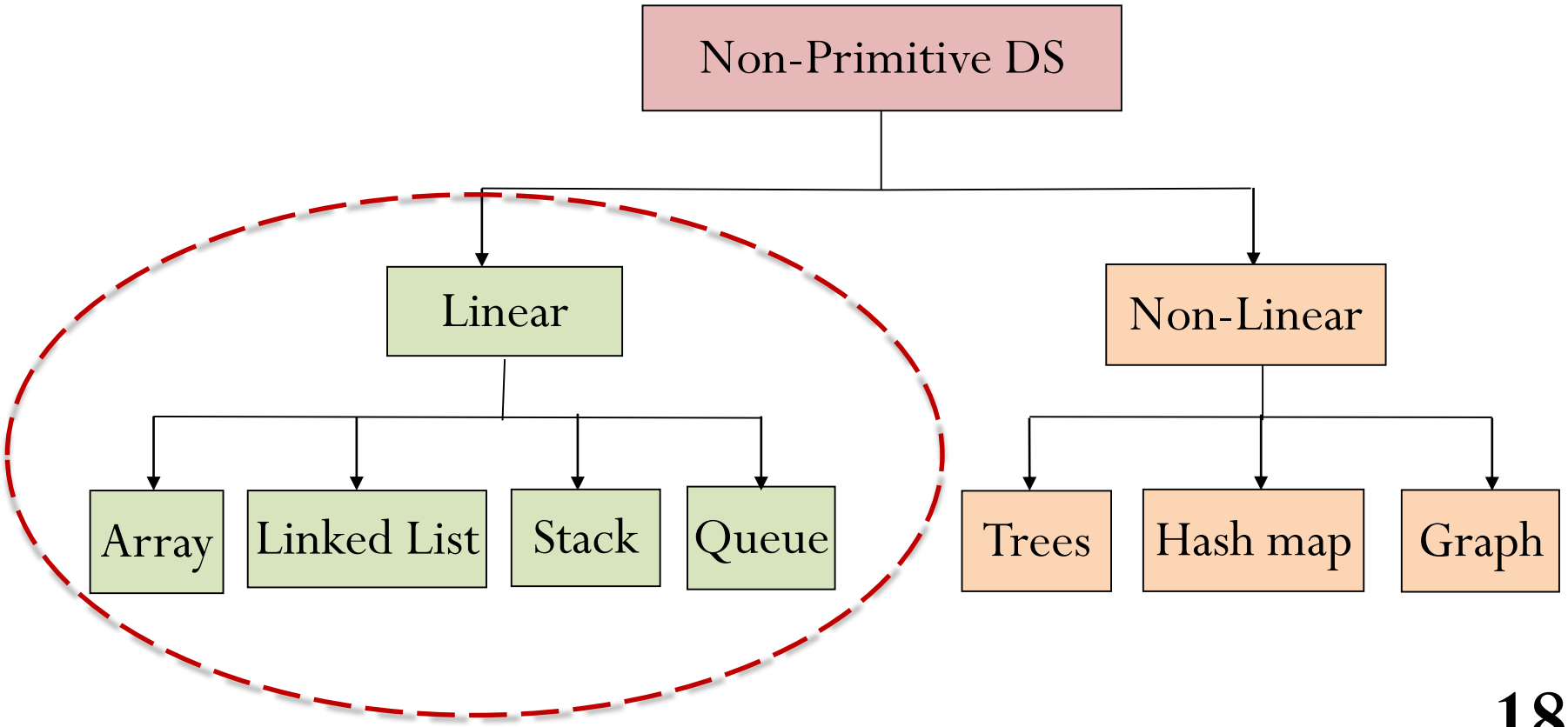
a) Primitive data structures

- **Basic structures**
  - Introduced in 1st year programming course(s)
  - Examples:
    - integers
    - characters
    - etc

b) Non-primitive data structures

- **More complex structures**
  - Derived from primitive data structures
- Different classifications:
  - Homogeneous vs heterogeneous – data type stored
  - Linear vs non-linear – arrangement of data items
  - Static vs dynamic – memory management (e.g. array vs linked list)
  - Etc.

- Further classification: linear vs. non-linear DSs



Common DS Operations

- Various activities can be performed on a DS:
  - Create a DS
  - Add/insert an item
  - Delete an item
  - Modify/change/update
  - Search - locate a specific item
  - Sort - order items
  - Merge
  - etc

Why study DS?

- Rem: Program processes data (I → P → O)
  - Data must be stored somewhere in memory
  - And MUST be organized properly
    - For **efficient** storage and retrieval
- Key qualities of the most appropriate (efficient) DS:
  - Offers easy storage, access and modification
  - Quick access to data
  - Uses less memory space.

Why study DS?

- Many DS application areas:
  - OS – process scheduling, e.g. printing (queues), dynamic memory allocation (linkedlists), etc
  - Graphics – arrays, LinkedLists, trees, etc
  - Simulation – queues, heaps, etc
  - Internet (e.g. stack for Back button, etc)
  - DBMS – indexing data items
  - Data analytics
  - AI and machine learning – quick retrieval of items
    - Arrays, LinkedLists, Hash Tables, Trees, etc
  - Etc

Summary

- Course objectives, overview
- Topics to cover in the course
- Course material
- Course admin: schedule, pre-requisite, labs
- Assessments
- Useful tools
- Introduction to Data Structures

Next Lesson

- Review: Java programming basics

Q & A